

Introduction to L^AT_EX

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Jackson Hole Classical Academy, 2026

```

1 \begin{document}
2   \selectlanguage{english}
3   \pagenumbering{gobble} % removes page numbers from the entire document
4
5   \noindent {\Large Introduction to \LaTeX} \vspace{18cm} \\
6   \noindent Elliott A. Best \\
7   \noindent \textit{Jackson Hole Classical Academy}, 2026 \vspace{2mm} \
  newline
8
9 \vspace*{3cm} We can add vertical space to the top of a page with a starred
  version of the command, and we can add vertical space after the first line
  without it. Space can be measured in millimeters, centimeters, and inches
  . It is good practice to close your paragraphs. \vspace{2cm} \par
10
11 If we can imagine it, we can create it in LaTeX. At first, LaTeX is far less
  intuitive than word processors like Microsoft Word. However, with a little
  effort you will quickly appreciate how easily it creates beautiful
  technical documents (like math homework) and how much power we have over
  every little detail (if we want it). \vspace{2mm} \par
12
13 We can make text \textbf{bold}, \textit{italicized}, \underline{underlined},
  or \textcolor{red}{colored}. In the preamble, you can see how to create
  custom colors with hex code. Note how ‘‘quotation marks’’ work. There are
  several better ways to add small horizontal spaces, but... \hspace{1cm}
  this works for any size space. You \, can \; also \quad try \quad these.
  \vspace{2mm} \par
14
15 \noindent This paragraph begins without an intent. This is a useful override
  for creating titles or emphasizing the beginning of a section of text. \
  vspace{2mm} \par
16
17 \begin{center}
18   Text can be centered in several ways, \\
19   but this is my favorite. \\
20   Note how to break the line \\
21   without starting a new paragraph.
22 \end{center} \vspace{2mm} \par
23
24 Many commands act on code in an environment created by a begin-end framework.
  Use indents in your code to add an extra dimension of visibility to these
  environments. I type almost all my code in Microsoft Word to increase the
  visibly even more. \vspace{2mm} \par
25
26 There are a few ways to change the size of our text. I usually just use
  commands like { \large this } or { \small this }. Percent signs make the
  compiler skip lines of text. We call these comments. Check out the
  comments here for more size commands. Notice how braces limit the effects
  of the commands above to just the words we want.
27
28 %For other font sizes, try tiny, small, large, Large, LARGE, huge, and Huge.
  For more customization, check out the anyfontsize package in the preamble.
29
30 \begin{comment} Here is a way to comment out large blocks of text. \end{
  comment}
31
32 \newpage

```

We can add vertical space to the top of a page with a starred version of the command, and we can add vertical space after the first line without it. Space can be measured in millimeters, centimeters, and inches. It is good practice to close your paragraphs.

If we can imagine it, we can create it in LaTeX. At first, LaTeX is far less intuitive than word processors like Microsoft Word. However, with a little effort you will quickly appreciate how easily it creates beautiful technical documents (like math homework) and how much power we have over every little detail (if we want it).

We can make text **bold**, *italicized*, underlined, or colored. In the preamble, you can see how to create custom colors with hex code. Note how “quotation marks” work. There are several better ways to add small horizontal spaces, but... this works for any size space. You can also try these.

This paragraph begins without an intent. This is a useful override for creating titles or emphasizing the beginning of a section of text.

Text can be centered in several ways,
but this is my favorite.
Note how to break the line
without starting a new paragraph.

Many commands act on code in an environment created by a begin-end framework. Use indents in your code to add an extra dimension of visibility to these environments. I type almost all my code in Microsoft Word to increase the visibly even more.

There are a few ways to change the size of our text. I usually just use commands like `this` or `this`. Percent signs make the compiler skip lines of text. We call these comments. Check out the comments here for more size commands. Notice how braces limit the effects of the commands above to just the words we want.

```

1 \begin{multicols}{2}
2   Here is some text split into two columns. We can change the column
   separation in the preamble. Here is some text split into two columns. You
   can change the column separation in the preamble. Here is some text split
   into two columns. You can change the column separation in the preamble.
   Here is some text split into two columns. You can change the column
   separation in the preamble. Here is some text split into two columns. You
   can change the column separation in the preamble.
3 \end{multicols} \vspace{5mm} % Note that the multicols function has built in
   vertical space.
4
5 \noindent This is how we create bullet points.
6 \begin{itemize}
7   \item First
8   \item Second
9   \item Third
10 \end{itemize}
11 \vspace{8mm}
12
13 \noindent This is how we create a list. The enumerate package has a built in
   outline feature, but you can make custom labels like I did here.
14 \begin{enumerate}
15   \item First
16   \item Second
17   \begin{enumerate}[label=\roman*.]
18     \item Something
19     \item Another thing
20   \end{enumerate}
21   \item Third
22 \end{enumerate}

```

Here is some text split into two columns. We can change the column separation in the preamble. Here is some text split into two columns. You can change the column separation in the preamble. Here is some text split into two columns. You

can change the column separation in the preamble. Here is some text split into two columns. You can change the column separation in the preamble. Here is some text split into two columns. You can change the column separation in the preamble.

This is how we create bullet points.

- First
- Second
- Third

This is how we create a list. The enumerate package has a built in outline feature, but you can make custom labels like I did here.

1. First
2. Second
 - i. Something
 - ii. Another thing
3. Third

```

1 When we write math expressions in a paragraph they look like this:  $\lim_{n \rightarrow 2} f(x)$ . But we can also force them to be as big as necessary to look
  ‘‘good’’:  $\displaystyle \lim_{n \rightarrow 2} f(x)$ . Sometimes we want to
  create a single line of emphasized mathematical text. Also, be sure to put
  subscripts and superscripts in braces if they are more than one character
  long.
2  $\int_{-1}^1 f(x)dx$ ;  $= \quad F(x) \big|_{-1}^1 = \quad F(1)-F(-1)$ 
3
4 Sometimes we want to create several lines of mathematical text that line up
  in a particular way. Note, if we take away the asterisk and simply use the
  align command, it will number the lines. I don’t usually find this
  helpful. In this function, the ampersand both creates a ‘‘box’’ (think:
  table) and defines the point of alignment. For example, this alignment
  environment has three columns (two ampersands per line) and one of those
  columns has special alignment at the equal sign (a third ampersand per
  line).
5
6 \begin{align*}
7   & \& (x+3)^2 \& =(x+3)(x+3) \\
8   & \text{Don't forget to break the line!} \& \& x^2+3x+3x+9 \\
9   & \& x^2+6x+9 \\
10  \end{align*}
11
12 \newpage

```

When we write math expressions in a paragraph they look like this: $\lim_{n \rightarrow 2} f(x)$. But we can also force them to be as big as necessary to look “good”: $\lim_{n \rightarrow 2} f(x)$. Sometimes we want to create a single line of emphasized mathematical text. Also, be sure to put subscripts and superscripts in braces if they are more than one character long.

$$\int_{-1}^1 f(x)dx = \left. F(x) \right|_{-1}^1 = F(1) - F(-1)$$

Sometimes we want to create several lines of mathematical text that line up in a particular way. Note, if we take away the asterisk and simply use the align command, it will number the lines. I don’t usually find this helpful. In this function, the ampersand both creates a “box” (think: table) and defines the point of alignment. For example, this alignment environment has three columns (two ampersands per line) and one of those columns has special a alignment at the equal sign (a third ampersand per line).

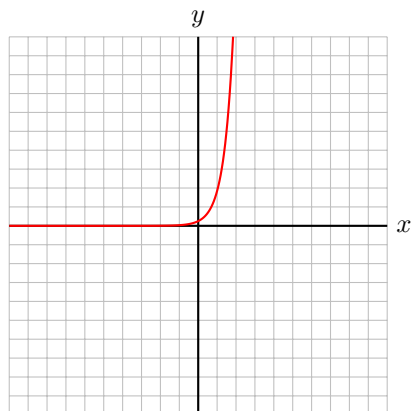
	$(x + 3)^2 = (x + 3)(x + 3)$
Don’t forget to break the line!	$= x^2 + 3x + 3x + 9$
	$= x^2 + 6x + 9$

```

1 \begin{flushleft}
2 \begin{minipage}{0.43\textwidth}
3   \noindent \textbf{Problem Number 3.} Here is a Cartesian coordinate
      system with a curve graphed onto it. Using the ‘‘minipage’’ feature, we
      can set it up to write some text with the graph on the side. Writing
      functions in this environment to create graphs can be really frustrating.
      Remember, we are telling the compiler to perform mathematical operations,
      not to display the functions.
4 \end{minipage}
5 \hfill
6 \begin{minipage}{0.45\textwidth}
7 \begin{center}
8   \begin{tikzpicture}[scale=0.25]
9     \draw[lightgray] (-10,-10) grid (10,10);
10
11     \coordinate[label=above:$y$](N) at (0,10);
12     \coordinate[label=right:$x$](K) at (10,0);
13     \coordinate(L) at (-10,0);
14     \coordinate(S) at (0,-10);
15
16     \draw[thick] (S)--(N);
17     \draw[thick] (L)--(K);
18
19     \draw[red,thick,smooth,samples=100,domain=-10:1.845] plot(\x,{(0.25)
      *(exp((2*\x)))});
20
21   \end{tikzpicture}
22 \end{center}
23 \end{minipage}
24 \end{flushleft}

```


Problem Number 3. Here is a Cartesian coordinate system with a curve graphed onto it. Using the “minipage” feature, we can set it up to write some text with the graph on the side. Writing functions in this environment to create graphs can be really frustrating. Remember, we are telling the compiler to perform mathematical operations, not to display the functions.

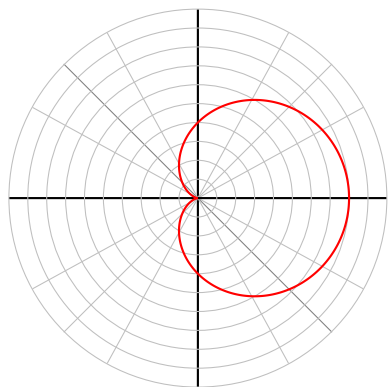


```

1 Here is a polar graph. Notice the subtle change in the function plots. The
  polar package uses a colon to separate the variables instead of a comma.
2 \vspace{5mm}
3
4 \begin{tikzpicture}[scale=0.25]
5
6     \coordinate(N) at (0,10);
7     \coordinate(K) at (10,0);
8     \coordinate(L) at (-10,0);
9     \coordinate(S) at (0,-10);
10
11     \coordinate(X) at (-7.1,-7.1);
12     \coordinate(Y) at (-7.1,7.1);
13     \coordinate(Z) at (7.1,-7.1);
14     \coordinate(P) at (7.1,7.1);
15
16     \coordinate(A) at (8.75,4.8);
17     \coordinate(B) at (-8.75,-4.8);
18     \coordinate(C) at (4.8,8.75);
19     \coordinate(D) at (-4.8,-8.75);
20     \coordinate(E) at (-8.75,4.8);
21     \coordinate(F) at (8.75,-4.8);
22     \coordinate(G) at (-4.8,8.75);
23     \coordinate(H) at (4.8,-8.75);
24
25     \draw[thick] (S)--(N);
26     \draw[color=gray] (Y)--(Z);
27     \draw[color=lightgray] (P)--(X);
28     \draw[color=lightgray] (A)--(B);
29     \draw[color=lightgray] (C)--(D);
30     \draw[color=lightgray] (E)--(F);
31     \draw[color=lightgray] (G)--(H);
32     \draw[thick] (L)--(K);
33
34     \draw[color=lightgray] (0,0) circle (1);
35     \draw[color=lightgray] (0,0) circle (2);
36     \draw[color=lightgray] (0,0) circle (3);
37     \draw[color=lightgray] (0,0) circle (4);
38     \draw[color=lightgray] (0,0) circle (5);
39     \draw[color=lightgray] (0,0) circle (6);
40     \draw[color=lightgray] (0,0) circle (7);
41     \draw[color=lightgray] (0,0) circle (8);
42     \draw[color=lightgray] (0,0) circle (9);
43     \draw[color=lightgray] (0,0) circle (10);
44
45     \draw[thick, red, domain=0:360,scale=4,samples=500] plot ({\x}:{(1)*(1+
      cos(\x))});
46
47 \end{tikzpicture}
48 \newpage

```

Here is a polar graph. Notice the subtle change in the function plots. The polar package uses a colon to separate the variables instead of a comma.

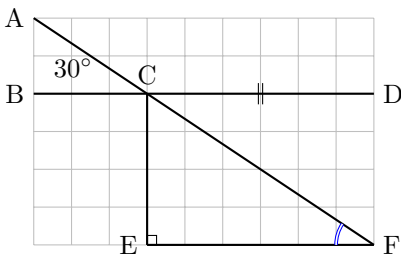


```

1 Here is a geometry diagram. I like to use a grid to create the diagram, and
  then either delete it or use comments to hide it. Note how we can move
  labels around, change the mark style, and scale labels and marks. Here, we
  are using the basic TikZ package (watch out for semicolons!) along with
  the TikZ Euclide package (no semicolons needed).
2 \vspace{2mm}
3
4 \begin{center}
5 \begin{tikzpicture}[scale=0.5]
6
7     \draw [lightgray] (0,0) grid (9,6);
8
9     \coordinate [label=left:A](A) at (0,6);
10    \coordinate [label=left:B](B) at (0,4);
11    \coordinate [label=above:C](C) at (3,4);
12    \coordinate [label=right:D](D) at (9,4);
13    \coordinate [label=left:E](E) at (3,0);
14    \coordinate [label=right:F](F) at (9,0);
15
16    \draw[thick] (D)--(B);
17    \draw[thick] (E)--(F);
18    \draw[thick] (A)--(F);
19    \draw[thick] (C)--(E);
20
21    \tkzLabelAngle[xshift=-5mm, yshift=2mm](A,C,B){$30^\circ$}
22    \tkzMarkRightAngle(F,E,C)
23    \tkzMarkSegment[mark=|](C,D)
24    \tkzMarkAngle[double, blue, scale=1](C,F,E)
25
26 \end{tikzpicture}
27 \end{center}

```

Here is a geometry diagram. I like to use a grid to create the diagram, and then either delete it or use comments to hide it. Note how we can move labels around, change the mark style, and scale labels and marks. Here, we are using the basic TikZ package (watch out for semicolons!) along with the TikZ Euclide package (no semicolons needed).



```

1 Here is a table. This is the best table package I've found. There are many.
   Inside the table are some useful commands. Note you can separate lines of
   code with empty comments to make your code easier to read without changing
   how the compiler reads it.
2
3 \begin{center}
4 \begin{tblr}{
5   colspec = {X[c,m]|X[c,m]},
6   rowsep = 6pt,
7   hlines = {1pt},
8   vlines = {1pt},}
9 Fractions &  $\frac{a}{b}$  \\
10 %
11 Braces &  $\{ a, b, c \}$  \\
12 %
13 Maps, Greek Letters &  $f:A \rightarrow B$ ,  $\alpha \mapsto \beta$  \\
14 %
15 Special L'hospital Command &  $\displaystyle \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)}$ 
    $\stackrel{\text{L'Hopital}}{=} \lim_{n \rightarrow \infty} \frac{f'(n)}{g'(n)}$  \\
16 %
17 \end{tblr}
18 \end{center}

```

Here is a table. This is the best table package I've found. There are many. Inside the table are some useful commands. Note you can separate lines of code with empty comments to make your code easier to read without changing how the compiler reads it.

Fractions	$\frac{a}{b}$
Braces	$\{a,b,c\}$
Maps, Greek Letters	$f : A \rightarrow B, \alpha \mapsto \beta$
Special L'hopital Command	$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} \stackrel{\text{L'H}}{=} \lim_{n \rightarrow \infty} \frac{f'(n)}{g'(n)}$